Fundamental Modules - 3

WEEK 3: CHASSIS, CONTROL & SAFETY

Days 13-18 | Detailed Hour-by-Hour Structure

Week Theme: *"Controlling the vehicle safely - suspension, steering, brakes, and safety systems"*

DAY 13: SUSPENSION SYSTEM + WHEEL HANDLING

Learning Objectives:

- Understand suspension purpose and types
- Identify suspension components
- Understand wheel alignment basics (awareness)
- Practice safe wheel handling and installation

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Suspension Fundamentals

Why Vehicles Need Suspension:

- Isolate occupants from road irregularities
- Maintain tire contact with road (traction)
- Control body movement (roll, pitch, dive)
- Provide comfortable ride
- Enable precise steering control

Sprung vs Unsprung Weight:

- **Sprung weight:** Supported by suspension (body, engine, passengers)
- **Unsprung weight:** Not supported (wheels, tires, brakes, hubs)
- Lower unsprung = better ride quality

Suspension Functions:

1. **Support vehicle weight**

2. **Absorb road shocks**
3. **Maintain tire contact**
4. **Minimize body movement**
5. **Provide steering control**

Suspension Motion:

- **Jounce (Compression):** Suspension compresses (wheel moves up)
- **Rebound (Extension):** Suspension extends (wheel moves down)
- **Spring stores energy, damper controls motion**

Ride vs Handling Trade-off:

- **Soft suspension:** Comfortable ride, more body roll
- **Stiff suspension:** Better handling, harsher ride
- Modern vehicles balance both

Hour 2 (10:00 AM - 11:00 AM): Suspension Components

Springs (Energy Storage):

1. Coil Springs:

- Most common type
- Helical steel spring
- Linear or progressive rate
- Compact, reliable
- Used on most modern vehicles

2. Leaf Springs:

- Multiple steel leaves stacked
- Traditional on trucks, SUVs
- Simple, robust
- Acts as locating arm also
- Heavy, less refined

3. Torsion Bars:

- Steel bar twists to store energy
- Common on older trucks
- Adjustable ride height
- Less common now

4. Air Springs:

- Air bags instead of steel
- Adjustable stiffness
- Used on luxury vehicles, buses
- Can level vehicle with load
- Complex, expensive

Shock Absorbers/Dampers:

- **Function:** Control spring oscillation
- **Without dampers:** Spring would bounce forever
- **With dampers:** Motion controlled, settles quickly

Types:

- **Twin-tube:** Traditional, two chambers
- **Monotube:** Single chamber, better performance
- **Gas-charged:** Nitrogen prevents foaming

Operation:

- Hydraulic fluid forced through valves
- Converts motion energy to heat
- Different valving for compression vs rebound
- Rebound typically stiffer than compression

Struts:

- Combines spring and damper in one unit
- Structural component (part of suspension)
- **MacPherson strut:** Most common front suspension
- More compact than separate spring/shock
- Includes upper mount with bearing (for steering)

Control Arms:

- Connect wheel hub to chassis
- Allow controlled up/down movement
- Provide attachment points for other components
- **Upper control arm:** Top of wheel (double wishbone)
- **Lower control arm:** Bottom of wheel (most designs)
- **Bushings:** Rubber isolators at pivot points
    - Allow movement, reduce noise

- Wear over time (check for cracks, tears)

Ball Joints:

- Connect control arms to wheel hub
- Allow steering and suspension movement
- Ball-and-socket design
- **Load-bearing:** Support vehicle weight
- **Follower:** Guide movement only
- Critical safety component (failure = wheel collapse)
- Grease-able (some) or sealed (most modern)

Sway Bar (Anti-Roll Bar):

- Reduces body roll in corners
- Connects left and right suspension
- When one side compresses, bar twists, lifts other side
- **Sway bar links:** Connect bar to control arms
- **Bushings:** Mount bar to chassis
- Common failure: worn links (clunking noise)

Hour 3 (11:00 AM - 12:00 PM): Suspension Types

Independent Suspension:

- Each wheel moves independently
- Better ride quality
- More complex
- Common on cars

1. MacPherson Strut:

- **Most common front suspension**
- **Components:** Strut, lower control arm, sway bar
- **Advantages:** Compact, simple, lightweight
- **Disadvantages:** Can't optimize geometry perfectly
- Used on: Most modern FWD cars

2. Double Wishbone (A-Arm):

- **Components:** Upper and lower control arms, separate shock/spring
- **Advantages:** Best handling, tunable geometry

- **Disadvantages:** More complex, takes more space
- Used on: Sports cars, luxury vehicles

3. Multi-Link:

- **Components:** 3-5 links per wheel
- **Advantages:** Excellent control, very tunable
- **Disadvantages:** Most complex, expensive
- Used on: High-end vehicles, rear suspension

Solid Axle (Beam Axle):

- Both wheels connected by solid beam
- Simple, robust
- **Advantages:** Strong, simple, cheap
- **Disadvantages:** Worse ride, less control
- Still used on: Trucks, off-road vehicles

Torsion Beam (Semi-Independent):

- Simpler than independent, better than solid
- Two trailing arms with torsion beam
- Common on small/medium car rear suspension
- Good balance of cost and performance

Suspension Geometry Terms (Awareness Level):

- **Camber:** Wheel tilt (top in/out)
- **Caster:** Steering axis tilt (forward/back)
- **Toe:** Wheel angle (front point in/out)
- These adjusted during wheel alignment
- Improper geometry = tire wear, poor handling

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Suspension Component Identification

Activity 1: Suspension Type Recognition (25 min)

- Identify suspension type on various vehicles:
  - Lift vehicle safely
  - Look at front suspension:
    - **MacPherson strut:** Single large strut unit visible

        - **Double wishbone:** Two control arms top and bottom
        - **Solid axle:** Beam connecting both wheels
    - Look at rear suspension:
        - **Independent:** Each wheel separate
        - **Torsion beam:** Connecting beam visible
        - **Solid axle:** Beam with differential

Activity 2: Component Location and Identification (35 min)

- Identify on actual vehicle:
    - **Springs:** Coil springs around struts or separate
    - **Shock absorbers/Struts:** Vertical damping units
    - **Control arms:** Lower arm (and upper if present)
    - **Ball joints:** Connection between arm and hub
    - **Bushings:** Rubber mounts at control arm pivots
    - **Sway bar:** Horizontal bar across vehicle
    - **Sway bar links:** Vertical links to control arms
    - **Sway bar bushings:** Hold bar to chassis
    - **CV axles (FWD):** Shafts with rubber boots
    - **Wheel bearings:** Inside hub (not visible, location only)
- Draw suspension system diagram
- Label all components
- Note suspension type (MacPherson, etc.)

Pass Criteria:

- ✅ Can identify suspension type
- ✅ Can locate all major suspension components
- ✅ Can distinguish springs from dampers
- ✅ Creates accurate suspension diagram

Hour 2 (2:00 PM - 3:00 PM): Suspension Visual Inspection

Activity 1: Shock/Strut Inspection (20 min)

- **Visual checks:**
    - Oil leaks on shock body (wetness = worn seal)
    - Dented or bent shock body
    - Damaged mounting bushings
    - Rust/corrosion

- **Bounce test:**
  - Push down firmly on each corner of vehicle
  - Release quickly
  - Should bounce once and settle
  - Multiple bounces = worn shocks
  - Do all four corners, compare

Activity 2: Spring and Control Arm Inspection (20 min)

- **Coil springs:**
  - Look for cracks (rare but serious)
  - Check for broken coils
  - Look for sagging (compare side-to-side)
  - Measure ride height if concerned
- **Control arms:**
  - Check bushings for cracks, tears
  - Look for bent arms (from impacts)
  - Check for excessive rust
  - Inspect ball joint boots (tears = contamination)

Activity 3: Sway Bar and Links Inspection (20 min)

- **Sway bar:**
  - Check bushings for wear, cracks
  - Should fit snugly
  - Worn = clunking over bumps
- **Sway bar links:**
  - Check ball joints at each end
  - Grasp and try to move (should be tight)
  - Look for torn boots
  - Listen for clicking/clunking
- **Common symptoms of worn components:**
  - Clunking over bumps = worn links, bushings
  - Bouncing, floating = worn shocks
  - Excessive body roll = worn sway bar bushings
  - Uneven tire wear = alignment, worn components

Pass Criteria:

- ✅ Performs bounce test correctly
- ✅ Inspects shocks for leaks
- ✅ Checks bushings for wear

- ✅ Identifies worn components
- ✅ Documents findings

Hour 3 (3:00 PM - 4:00 PM): Wheel Removal Practice & Torque

Activity 1: Proper Wheel Removal Technique Review (15 min)

- Review from Day 1, with added focus:
  1. Loosen lug nuts 1/4 turn (vehicle on ground)
  2. Lift vehicle safely
  3. Remove lug nuts completely
  4. Pull wheel straight off hub
  5. **Inspect hub surface** (rust, damage)
  6. **Inspect wheel mating surface** (clean)
  7. **Check lug nuts** (worn, damaged threads)

Activity 2: Proper Wheel Installation Sequence (25 min)

- **Critical procedure:**
  1. Clean hub surface (wire brush if rusty)
  2. Check wheel center hole matches hub pilot diameter
  3. Align wheel with studs/holes
  4. Start all lug nuts by hand (cross-threading check)
  5. Snug lug nuts in star pattern (hand tight with wrench)
  6. Lower vehicle until tire touches ground (weight on wheel)
  7. **Final torque in star pattern with torque wrench**
- **Star pattern importance (5-lug example):**

```
      1
  5         2
    3     4
```

- Tighten: 1-3-5-2-4
- Prevents warping brake rotors
- Ensures even clamping force
- **Torque specifications:**
  - Check door jamb sticker or manual
  - Common: 80-110 Nm (varies by vehicle)
  - Aluminum wheels often lower than steel
  - **Must use torque wrench for final tightening!**

Activity 3: Torque Wrench Practice (20 min)

- Each student practices:
  - Setting torque wrench to specification
  - Proper torque sequence (star pattern)
  - Feeling the "click" at correct torque
  - Not continuing to pull after click
  - Verifying all lug nuts torqued
- **Common mistakes to avoid:**
  - Tightening in circle pattern (warps rotor)
  - Over-torquing (strips threads, warps rotor)
  - Under-torquing (wheel can loosen)
  - Not using torque wrench at all
  - Using impact wrench for final tightening
- **Re-torque reminder:**
  - After 50-100 km, re-check torque
  - Especially after tire change
  - Wheels can settle, lug nuts can loosen slightly
  - Good practice = customer safety

Pass Criteria:

- ✅ Cleans hub surfaces before installation
- ✅ Starts all lug nuts by hand
- ✅ Uses correct star pattern
- ✅ Torques to specification correctly
- ✅ Verifies all lug nuts properly torqued

DAY 13 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies suspension type correctly
- [ ] Locates all suspension components
- [ ] Performs bounce test
- [ ] Inspects shocks/struts for leaks
- [ ] Checks bushings and links
- [ ] Removes and installs wheel correctly
- [ ] Torques lug nuts with proper sequence

Knowledge Check (Verbal):

- What are the two main functions of suspension?
- What is the difference between a spring and a shock absorber?
- What is the most common front suspension type?
- Why must we torque lug nuts in a star pattern?
- What does a bounce test check?

DAY 14: STEERING SYSTEM + ALIGNMENT AWARENESS

Learning Objectives:

- Understand steering system types and operation
- Identify steering components
- Understand wheel alignment principles (awareness level)
- Check steering operation and play

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Steering Fundamentals

Steering System Purpose:

- Control vehicle direction
- Provide feedback to driver
- Multiply driver force (mechanical advantage)
- Return to center after turn
- Maintain stable straight-line tracking

Steering Ratio:

- Relationship between steering wheel rotation and wheel angle
- Example: 16:1 ratio
  - Steering wheel turns 16 degrees
  - Front wheels turn 1 degree
- **Higher ratio (18:1, 20:1):**
  - Easier to turn
  - Less feedback
  - More steering wheel rotation needed
  - Used on trucks, comfort vehicles

- **Lower ratio (12:1, 14:1):**
  - More effort required
  - Better feedback
  - Quicker response
  - Used on sports cars

Steering Geometry:

- **Ackermann steering:**
  - Inside wheel turns sharper than outside
  - Prevents tire scrubbing
  - Achieved through steering arm angles
- **Steering axis:**
  - Imaginary line through upper and lower ball joints (or strut mount)
  - Determines steering behavior

Hour 2 (10:00 AM - 11:00 AM): Steering System Types

Manual Steering (Rare on Modern Vehicles):

- Driver effort only
- Simple, reliable
- Heavy at low speeds
- Light weight, no parasitic loss
- Used on small, light vehicles only

Power Steering Systems:

1. Hydraulic Power Steering (Traditional):

- **Components:**
  - **Hydraulic pump:** Belt-driven by engine
  - **Power steering fluid reservoir**
  - **High-pressure hoses**
  - **Steering rack/gearbox:** Contains hydraulic assist
  - **Return hose:** Low pressure back to reservoir
- **Operation:**
  - Pump runs whenever engine runs
  - Creates 50-100 bar pressure
  - When steering turned, valve directs pressure to assist
  - Reduces driver effort by 80-90%

- **Advantages:**
  - Good road feel
  - Powerful assist
  - Proven technology
- **Disadvantages:**
  - Parasitic loss (pump always running)
  - Fluid maintenance required
  - Can leak

2. Electric Power Steering (EPS - Modern):

- **Components:**
  - **Electric motor:** Provides assist
  - **Torque sensor:** Measures driver input
  - **ECU:** Controls motor
  - **Position sensors:** Know steering angle
- **Types:**
  - **Column-assist:** Motor on steering column
  - **Pinion-assist:** Motor on steering rack pinion
  - **Rack-assist:** Motor on rack itself (most powerful)
- **Operation:**
  - ECU reads torque sensor (how hard driver turning)
  - ECU commands motor to assist proportionally
  - Variable assist (speed-dependent):
    - More assist at parking speeds
    - Less assist at highway speeds
  - Motor only works when needed (not constantly)
- **Advantages:**
  - No parasitic loss when not steering
  - Better fuel economy (2-3%)
  - No fluid, no leaks, no maintenance
  - Programmable assist (sport modes, etc.)
  - Lighter weight
- **Disadvantages:**
  - Sometimes less road feel
  - More expensive
  - Complex electronics

Hour 3 (11:00 AM - 12:00 PM): Steering Components

Steering Wheel:

- Driver input point
- Contains airbag (safety)
- May have controls (cruise, audio, phone)
- Should have 0-2 cm free play

Steering Column:

- Connects wheel to rack/gearbox
- **Collapsible:** Safety feature (collapses in crash)
- **Adjustable:** Tilt and telescoping (many vehicles)
- **Universal joints:** Allow angle changes

Rack and Pinion Steering (Most Modern Vehicles):

- **Pinion gear:** Attached to steering column
- **Rack:** Horizontal bar with teeth
- Pinion rotates, moves rack side-to-side
- **Tie rods:** Connect rack ends to wheel hubs
- Compact, efficient, direct feel
- Used on almost all cars

Recirculating Ball Steering (Older Vehicles, Trucks):

- **Worm gear:** On steering column
- **Recirculating ball nut:** Rides on worm
- **Sector shaft:** Converts to rotary motion
- **Pitman arm:** Transfers motion to linkage
- More durable for heavy-duty
- Less precise feedback
- Takes more space

Steering Linkage Components:

Tie Rods:

- Connect steering rack to wheel hubs
- **Inner tie rod:** Connects to rack (covered by boot)
- **Outer tie rod:** Connects to hub (threaded adjustment)
- **Tie rod boot:** Rubber cover protecting inner ball joint
- Adjustment changes toe alignment
- Critical safety component

Steering Knuckle:

- Wheel hub mounts to this
- Connects to suspension (ball joints/strut)
- Steering arm for tie rod connection
- Brake caliper mounts here

Steering Gear Boots:

- Rubber covers on rack ends
- Protect inner tie rods
- Keep dirt/water out
- Tears = contamination = premature wear

Power Steering Fluid (Hydraulic Systems):

- ATF or special PS fluid
- Functions:
  - Hydraulic medium
  - Lubricates pump and rack
  - Cools system
- Check level regularly
- Change if contaminated

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Steering System Identification

Activity 1: Steering Type Recognition (20 min)

- Determine steering type on vehicles: **Hydraulic Power Steering:**
  - Look for power steering fluid reservoir (near engine)
  - Look for hydraulic pump (belt-driven, at front of engine)
  - Look for high-pressure hoses (metal lines with rubber hoses)
  - Fluid dipstick or level marks on reservoir

  Electric Power Steering:
  - No fluid reservoir
  - No hydraulic pump
  - Look for electric motor on steering column or rack
  - Often has electrical connectors near steering area

  Manual Steering (rare):
  - No reservoir

    - No pump
    - Simple, direct mechanical connection

Activity 2: Steering Component Location (40 min)

- Identify components on actual vehicle:
    - **Steering wheel:** Driver input
    - **Steering column:** Under dashboard to firewall
    - **U-joints:** Where column angles
    - **Steering rack:** Behind engine/under firewall (may need to lift)
    - **Rack boots:** Rubber covers on each end
    - **Inner tie rods:** Inside boots (not visible without removal)
    - **Outer tie rods:** Visible connections to wheel hubs
    - **Tie rod ends:** Ball joints at hub connection
    - **Steering knuckle/hub:** Where wheel attaches
- If hydraulic system present:
    - **Power steering pump:** Front of engine
    - **Reservoir:** Plastic tank with fluid
    - **High-pressure hose:** From pump to rack
    - **Return hose:** From rack back to reservoir
- Draw steering system diagram
- Trace steering input from wheel to tire
- Label all components

Pass Criteria:

- ✅ Can identify steering system type
- ✅ Can locate steering rack
- ✅ Can identify tie rods and boots
- ✅ Can locate PS pump/motor (if present)
- ✅ Creates accurate steering diagram

Hour 2 (2:00 PM - 3:00 PM): Steering Inspection & Play Check

Activity 1: Power Steering Fluid Check (15 min)

- **For hydraulic PS systems:**
    1. Locate reservoir
    2. Check level marks (MIN/MAX)
    3. **Some checked HOT, some COLD** (check cap marking)
    4. Check fluid color:

        - New: Red or clear
        - Used: Brownish
        - Bad: Black, burnt smell
    5. Check for leaks at pump, hoses, rack
    6. Top up if needed (correct fluid type)
- **Common leak locations:**
    - Pump shaft seal
    - High-pressure hose connections
    - Rack end seals (inside boots)
    - Reservoir cap (pressure venting)

Activity 2: Steering Play Check (20 min)

- **Free play measurement:**
    - Wheels straight ahead
    - Engine OFF (no power assist)
    - Gently turn steering wheel left and right
    - Measure movement before wheels move
    - **Should be < 5cm at wheel rim**
    - Excessive play = worn components
- **With engine running (PS equipped):**
    - Turn wheel lock-to-lock (full left to full right)
    - Should be smooth, no binding
    - Should return to center when released
    - Listen for noises:
        - Groaning = low fluid or failing pump
        - Whining = air in system or worn pump
        - Clicking = failing CV joint or tie rod

Activity 3: Visual Steering Inspection (25 min)

- Lift vehicle safely
- Inspect steering components: **Tie rod boots:**
    - Check for tears, cracks
    - Feel for grease (indicates boot failure)
    - Torn boot = contamination = premature wear

  Tie rod ends:
    - Grasp wheel at 3 and 9 o'clock
    - Try to move in/out (push-pull)
    - Should be NO movement at tie rod
    - Movement = worn ball joint (replace)

Rack boots:
  - Squeeze boots (should feel springy, full)
  - Collapsed, flat boot = leaking rack
  - Look for fluid/grease escaping

Steering rack:
  - Look for fluid leaks
  - Check mounting bushings
  - Ensure secure to chassis

Power steering hoses (if equipped):
  - Check for leaks at connections
  - Check for cracks in rubber hoses
  - Check for chafing against other parts
- Document all findings
- Create inspection report

Pass Criteria:

- ✅ Checks PS fluid level (if equipped)
- ✅ Measures steering free play correctly
- ✅ Inspects tie rod boots
- ✅ Checks for play in tie rod ends
- ✅ Identifies worn or damaged components

Hour 3 (3:00 PM - 4:00 PM): Wheel Alignment Awareness

Activity 1: Alignment Basics Understanding (25 min)

- **What is wheel alignment?**
  - Adjusting angles of wheels relative to each other and ground
  - Ensures:
    - Straight-line tracking
    - Even tire wear
    - Proper handling
    - Reduced fuel consumption

Three Main Angles:

1. Camber:

- Tilt of wheel viewed from front
- **Positive camber:** Top of tire tilts outward

- **Negative camber:** Top of tire tilts inward
- **Zero camber:** Perfectly vertical
- **Effect if wrong:**
  - Positive = outside edge wear
  - Negative = inside edge wear
  - Vehicle pulls to side with more positive camber

2. Caster:

- Tilt of steering axis viewed from side
- **Positive caster:** Axis tilts toward rear at top
  - Most vehicles have positive caster
  - Provides self-centering
  - Improves high-speed stability
- **Effect if wrong:**
  - Vehicle wanders
  - Poor self-centering
  - Uneven caster = pulls to side with less caster

3. Toe:

- Wheel angle viewed from above
- **Toe-in:** Front of tires point toward each other
- **Toe-out:** Front of tires point away from each other
- **Effect if wrong:**
  - Toe-in = feathering on outside edges
  - Toe-out = feathering on inside edges
  - Fastest tire wear if wrong
  - Most commonly adjusted angle

Activity 2: Alignment Problem Symptoms (20 min)

- **Symptoms requiring alignment:**
  - Vehicle pulls to one side
  - Steering wheel off-center when driving straight
  - Uneven tire wear (inside or outside edges)
  - Steering wheel doesn't return to center
  - Vehicle wanders, needs constant correction
  - Tire squealing in turns
- **What causes misalignment:**
  - Hitting potholes, curbs
  - Minor accidents

    - Worn suspension components
    - Lowering/lifting vehicle
    - Normal wear over time
- **When to check alignment:**
    - After replacing suspension components
    - After tire replacement (if wear was uneven)
    - If vehicle pulls or steering off-center
    - Annually as preventive maintenance
    - After any front-end collision

NOTE: Students are NOT performing alignments (requires special equipment)

- Alignment = specialization track skill
- Understanding symptoms = sufficient for fundamentals

Activity 3: Tire Wear Pattern Analysis (15 min)

- **Identifying wear patterns: Even wear across tread:**
    - Normal, proper inflation and alignment

  Center wear:
    - Over-inflation
    - Too much pressure, only center contacts road

  Edge wear (both sides):
    - Under-inflation
    - Not enough pressure, only edges contact

  Inside OR outside edge wear:
    - Camber misalignment
    - Inside wear = negative camber
    - Outside wear = positive camber

  Feathering (one side of tread blocks worn):
    - Toe misalignment
    - Tire being "scrubbed" sideways while rolling

  Cupping (scalloped dips):
    - Worn suspension components (shocks)
    - Wheel balance issues
    - Tire defect

  Flat spots:
    - Emergency braking (locked wheels)
    - Vehicle parked long time same spot
- Students examine actual tires
- Identify wear patterns
- Determine likely cause

- Create wear analysis report

Pass Criteria:

- ✅ Can explain three main alignment angles
- ✅ Recognizes symptoms of misalignment
- ✅ Can identify tire wear patterns
- ✅ Can suggest likely causes of wear
- ✅ Understands when alignment needed

DAY 14 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies steering system type
- [ ] Locates all steering components
- [ ] Checks PS fluid level (if equipped)
- [ ] Measures steering free play
- [ ] Inspects tie rod boots and ends
- [ ] Identifies tire wear patterns
- [ ] Recognizes symptoms of misalignment

Knowledge Check (Verbal):

- What are the two main types of power steering?
- Name three main steering components
- What is acceptable steering wheel free play?
- What does excessive toe-out cause?
- What causes inside edge tire wear?

DAY 15: BRAKE SYSTEMS + BRAKE INSPECTION

Learning Objectives:

- Understand hydraulic brake principles
- Identify brake system components
- Understand ABS basics
- Perform brake inspection and measurements

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Brake System Fundamentals

Why Brakes Are Critical:

- **Most important safety system**
- Convert kinetic energy to heat energy
- Must work perfectly every time
- Subject to extreme conditions
- Regular inspection essential

Hydraulic Brake Principles:

- Based on Pascal's Law: Pressure in fluid is equal throughout
- **Force multiplication:**
  - Small force on small piston (brake pedal)
  - Creates large force on large piston (wheel brake)
  - Typical ratio: 4:1 to 6:1 multiplification

Brake System Requirements:

- Must stop vehicle safely in all conditions
- Must be powerful enough for vehicle weight
- Must not fade with repeated use
- Must work even if one circuit fails (dual circuit)
- Must be easy to modulate (driver control)

Brake Friction:

- Friction pads/shoes pressed against rotating disc/drum
- Kinetic energy → Heat energy
- Heat must be dissipated (cooling)
- Coefficient of friction determines stopping power
- More friction = shorter stopping distance

Brake Heat:

- Normal braking: 200-300°C
- Hard braking: 500-700°C
- Extreme: 1000°C+ (racing, mountain descents)
- Too much heat:
  - Brake fade (reduced friction)
  - Boiling brake fluid (vapor in lines = spongy pedal)

    - Warped rotors
    - Glazed pads

Hour 2 (10:00 AM - 11:00 AM): Brake System Components

Master Cylinder:

- **Function:** Convert pedal force to hydraulic pressure
- **Components:**
    - Reservoir (stores brake fluid)
    - Primary piston (front brakes)
    - Secondary piston (rear brakes)
    - Return springs
    - Seals
- **Dual circuit:** Two independent hydraulic circuits
    - Front and rear separate
    - If one circuit fails, other still works
    - Safety requirement
- **Location:** Firewall, near brake pedal
- **Common failures:**
    - Internal seal leakage (pedal sinks slowly)
    - External leakage (fluid drips)
    - Corrosion

Brake Lines and Hoses:

- **Steel brake lines:**
    - Rigid pipes run length of vehicle
    - Must be corrosion-resistant
    - Can rust through (dangerous!)
- **Flexible brake hoses:**
    - At each wheel (allows suspension movement)
    - Rubber with braided reinforcement
    - Can crack, bulge, leak
- **Brake fluid:** DOT 3, DOT 4, DOT 5.1
    - Hydraulic medium
    - High boiling point
    - Hygroscopic (absorbs moisture over time)
    - Moisture lowers boiling point = fade
    - Must be changed every 2-3 years

Disc Brakes (Front - Always; Rear - Most Modern):

- **Components:**
  - **Brake rotor (disc):** Cast iron disc bolted to hub
    - Rotates with wheel
    - Smooth friction surface
    - Vented (internal cooling passages) or solid
  - **Brake caliper:** Clamps rotor
    - Fixed caliper: Pistons both sides (performance)
    - Floating caliper: Piston one side, slides on pins (common)
  - **Brake pads:** Friction material
    - Backing plate (steel)
    - Friction material (organic, semi-metallic, ceramic)
    - Wear indicator (metal tab squeals when worn)
  - **Caliper pistons:** Push pads against rotor
  - **Caliper slides/pins:** Allow floating caliper to move
- **Operation:**
  1. Press brake pedal
  2. Master cylinder creates pressure
  3. Pressure forces caliper piston(s) out
  4. Pistons push pads against rotor
  5. Friction slows rotor (and wheel)
  6. Release pedal
  7. Caliper seals retract pistons slightly
  8. Pads barely clear rotor
- **Advantages:**
  - Better heat dissipation (rotor exposed to air)
  - Less fade
  - Self-cleaning (centrifugal force throws water/dirt off)
  - Easier to inspect and service
  - Better performance

Drum Brakes (Rear - Some Vehicles):

- **Components:**
  - **Brake drum:** Rotating cylinder bolted to hub
  - **Brake shoes:** Curved friction material (two per wheel)
  - **Wheel cylinder:** Contains two pistons
  - **Return springs:** Pull shoes away from drum
  - **Adjuster:** Takes up clearance as shoes wear
  - **Backing plate:** Stationary, shoes mount here

- **Operation:**
  1. Press brake pedal
  2. Pressure enters wheel cylinder
  3. Pistons push shoes outward against drum
  4. Friction slows drum (and wheel)
  5. Release pedal
  6. Springs pull shoes back
- **Advantages:**
  - Cheaper to manufacture
  - Integrated parking brake
  - Good for low-load applications (rear brakes)
- **Disadvantages:**
  - Poor heat dissipation (enclosed design)
  - More fade
  - Harder to inspect (must remove drum)
  - More complex adjustment

Parking Brake (Emergency Brake):

- Mechanical system (cables, not hydraulic)
- **Disc brake vehicles:** Separate mechanism in rear caliper OR small drum inside rear rotor
- **Drum brake vehicles:** Uses same shoes, mechanical actuation
- Must hold vehicle on steep hill
- Should be applied every time vehicle parked
- Cable can stretch, requires adjustment

Hour 3 (11:00 AM - 12:00 PM): ABS & Advanced Brake Systems

Anti-Lock Braking System (ABS):

- **Purpose:** Prevent wheel lockup during hard braking
- **Why it matters:**
  - Locked wheel = no steering control
  - Locked wheel = longer stopping distance (less friction)
  - ABS maintains traction and steering control

ABS Components:

- **Wheel speed sensors:** Monitor each wheel's speed
- **Hydraulic modulator:** Valves control pressure to each wheel

- **ABS pump:** Restores pressure quickly
- **ABS ECU:** Computer controls system
- **ABS warning light:** Dashboard indicator

ABS Operation:

1. Driver brakes hard
2. Wheel speed sensors detect wheel about to lock
3. ABS ECU commands modulator to reduce pressure to that wheel
4. Wheel speed increases (stops skidding)
5. ABS restores pressure
6. Cycle repeats many times per second (5-15 Hz)
7. Driver feels pulsing in pedal (normal!)

Using ABS Correctly:

- **In emergency:** Press brake pedal HARD and HOLD
- DO NOT pump brakes (ABS does this for you)
- DO NOT release pedal (even with pulsing)
- Keep steering control (ABS allows this)
- ABS increases stopping distance on gravel/snow (but maintains control)

ABS Limitations:

- Can't overcome physics (ice is slippery)
- Increases stopping distance on loose surfaces
- Requires all wheels same size tires
- Sensor failures can disable system

Electronic Brake Distribution (EBD):

- Integrated with ABS
- Automatically adjusts front/rear brake balance
- Accounts for load (passengers, cargo)
- Prevents rear lockup during hard braking

Brake Assist (BA/BAS):

- Detects emergency braking (rapid pedal application)
- Applies maximum braking force automatically
- Compensates for drivers not pressing hard enough
- Common on modern vehicles

Hill Hold Control:

- Prevents rollback on hills
- Holds brakes for 2-3 seconds after driver releases
- Allows time to move foot from brake to accelerator
- Useful on steep hills

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Brake System Identification

Activity 1: Brake Type Recognition (20 min)

- Identify brake type at each wheel:
  - **Remove wheel** (already practiced)
  - **Front:** Look behind wheel
    - Disc brake: Shiny rotor visible, caliper on top or behind
  - **Rear:** Look behind wheel
    - Disc brake: Same as front
    - Drum brake: Large cylinder (drum) visible, no rotor
- Identify brake configuration:
  - "4-wheel disc" = discs on all four wheels
  - "Disc front, drum rear" = common on economy cars

Activity 2: Disc Brake Component Identification (40 min)

- With wheel removed, identify:
  - **Brake rotor:**
    - Flat friction surface (should be smooth)
    - Vented (cooling passages) or solid
    - Minimum thickness marking on edge
  - **Brake caliper:**
    - Clamps over rotor
    - Fixed or floating (most are floating)
  - **Brake pads:**
    - Visible through caliper opening
    - Friction material thickness
    - Backing plate
  - **Caliper piston(s):**
    - Inside caliper (push pads)
  - **Caliper bracket:**
    - Bolted to hub

        - Caliper slides on this (floating calipers)
    - **Slide pins:**
        - Allow caliper to float
        - Must move freely
    - **Brake hose:**
        - Flexible hose from chassis to caliper
        - Allows suspension movement
- If drum brake accessible:
    - **Brake drum:** Remove to see internal components
    - **Brake shoes:** Two curved shoes
    - **Wheel cylinder:** At top, contains pistons
    - **Return springs:** Pull shoes back
    - **Adjuster:** Star wheel or automatic type
- Draw brake system diagram (disc or drum)
- Label all components

Pass Criteria:

- ✅ Can identify disc vs drum brakes
- ✅ Can locate all disc brake components
- ✅ Can identify caliper type (fixed/floating)
- ✅ Creates accurate brake diagram

Hour 2 (2:00 PM - 3:00 PM): Brake Inspection & Measurement

Activity 1: Brake Pad Thickness Measurement (25 min)

- **Why pad thickness matters:**
    - Pads wear with use
    - Too thin = metal-to-metal (rotor damage)
    - Minimum: 3mm (replace before this)
    - New: 10-12mm typical
- **Measurement procedure:**
    1. Look through caliper opening
    2. Locate friction material (not backing plate)
    3. Measure thinnest point with caliper or ruler
    4. Check both pads (inner and outer)
    5. Record thickness
    6. Compare to minimum (3mm)
    7. Check all four wheels

- **Pad wear indicators:**
  - Metal tab on pad
  - When pad wears to ~2mm, tab contacts rotor
  - Creates squealing noise (warning to replace)
  - If hearing squeal, pads are at replacement point
- **Uneven pad wear:**
  - Inner pad thinner than outer = stuck caliper slide
  - Outer pad thinner than inner = rare, check bracket
  - Tapered wear = caliper alignment issue

Activity 2: Brake Rotor Inspection (20 min)

- **Visual inspection:**
  - **Surface condition:**
    - Smooth = good
    - Light grooves = normal wear
    - Deep grooves = needs resurfacing or replacement
    - Cracks = replace immediately (safety!)
    - Rust on friction surface = surface rust OK (will clean off), deep rust = inspect further
  - **Rotor thickness:**
    - Find minimum thickness stamped on rotor (often on outer edge)
    - Measure actual thickness with micrometer (if available)
    - Must be above minimum
    - Too thin = can't dissipate heat, dangerous
  - **Warped rotor:**
    - Spin rotor by hand
    - Watch edge relative to fixed point
    - Should spin true (no wobble)
    - Wobbling = warped (steering wheel vibration when braking)
    - Causes: Excessive heat, over-torqued lug nuts

Activity 3: Brake System Component Check (15 min)

- **Brake hoses:**
  - Check for cracks, bulges, leaks
  - Should be flexible, not stiff
  - No fluid on outside
  - Connections tight
- **Caliper slide pins:**
  - Should move freely in and out

    - Stuck = uneven pad wear
    - Clean and lube pins during pad service
- **Brake fluid level:**
    - Already checked on Day 7
    - Level drops as pads wear (normal)
    - Sudden drop = leak (check immediately)
- **ABS wheel speed sensors:**
    - Located at each wheel hub
    - Wire connection to sensor
    - Should not be damaged
    - Gap to tone ring critical (about 1mm)
- Document findings:
    - Pad thickness (all four wheels)
    - Rotor condition
    - Any issues found

Pass Criteria:

- ✅ Can measure brake pad thickness
- ✅ Can inspect rotor condition
- ✅ Recognizes when pads need replacement
- ✅ Checks brake hoses and components
- ✅ Documents brake system condition

Hour 3 (3:00 PM - 4:00 PM): Brake Problem Diagnosis

Activity 1: Common Brake Problems (Discussion) (25 min)

Problem 1: Squealing Brakes

- **Causes:**
    - Wear indicator contacting rotor (pads worn out)
    - Cheap/low-quality pads
    - Glazed pads (overheated)
    - Missing anti-squeal shims
    - Rust on backing plate
    - Normal (some pad materials squeal slightly)

Problem 2: Grinding Noise

- **Causes:**

  - Pads completely worn (metal-to-metal)
  - Rust grooves in rotor
  - Foreign material between pad and rotor
  - **Requires immediate attention!**

Problem 3: Pulsating Brake Pedal

- **Causes:**
  - Warped rotor (most common)
  - Loose wheel bearing
  - Loose caliper mounting
  - Variation in rotor thickness
- **Symptom:** Pedal pulses up/down during braking

Problem 4: Brake Pedal Goes to Floor

- **Causes:**
  - Brake fluid leak (external)
  - Master cylinder internal leak
  - Air in brake lines
  - **Very dangerous - do not drive!**

Problem 5: Spongy Brake Pedal

- **Causes:**
  - Air in brake lines
  - Old brake fluid (moisture boiling)
  - Flexible brake hose bulging
  - Requires bleeding

Problem 6: Vehicle Pulls to One Side

- **Causes:**
  - Stuck caliper on one side
  - Contaminated pad (oil, brake fluid)
  - Uneven pad wear
  - Tire pressure difference (check first!)

Problem 7: ABS Light On

- **Causes:**
  - Wheel speed sensor failure
  - Wiring damage
  - ABS module failure

    - Low brake fluid (some systems)
- **Note:** Brakes still work, but no ABS function

Activity 2: Brake Problem Scenarios (20 min)

- **Roleplay practice:**
    - Instructor describes customer complaint
    - Students diagnose likely cause
    - Students explain to "customer"
- **Scenarios:**
    - "My brakes make a squealing noise"
    - "The brake pedal feels soft and spongy"
    - "The steering wheel shakes when I brake"
    - "My car pulls to the right when braking"
    - "I hear grinding when I brake"

Activity 3: Parking Brake Check (15 min)

- **Parking brake test:**
    1. On level ground
    2. Apply parking brake fully
    3. Try to drive (should not move)
    4. If moves = needs adjustment or worn
- **Parking brake mechanism:**
    - Lever or pedal in cabin
    - Cables run to rear brakes
    - Mechanical actuation (not hydraulic)
    - Can stretch over time
- **Adjustment check:**
    - Count clicks when applying
    - Typical: 5-8 clicks to fully engaged
    - Too many clicks (>10) = needs adjustment
    - Too few clicks (<3) = over-adjusted

Pass Criteria:

- ✅ Can identify symptoms of brake problems
- ✅ Can suggest likely causes
- ✅ Can explain problems to customers
- ✅ Can test parking brake function
- ✅ Understands when immediate service required

DAY 15 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies disc and drum brakes
- [ ] Locates brake system components
- [ ] Measures brake pad thickness
- [ ] Inspects brake rotors
- [ ] Checks brake hoses and lines
- [ ] Recognizes brake wear indicators
- [ ] Tests parking brake
- [ ] Diagnoses common brake problems

Knowledge Check (Verbal):

- Why is brake fluid hygroscopic a problem?
- What is minimum brake pad thickness?
- What causes brake pedal pulsation?
- How does ABS work?
- What should you do if brake pedal goes to floor?

DAY 16: TYRES & WHEELS + BALANCING AWARENESS

Learning Objectives:

- Understand tyre construction and types
- Read tyre sidewall information
- Perform tyre inspection and tread depth measurement
- Understand wheel balancing principles

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Tyre Fundamentals

Why Tyres Are Critical:

- **Only contact point with road** (about credit card size per tyre)
- Affect:
  - Safety (braking, cornering, stability)
  - Comfort (absorb road irregularities)

- Fuel economy (rolling resistance)
- Handling and performance
- Noise

Tyre Construction:

Basic Components:

1. **Tread:** Outer rubber with pattern
   - Contacts road surface
   - Provides grip (dry and wet)
   - Disperses water (prevents hydroplaning)
   - Compound affects wear vs grip
2. **Tread pattern:**
   - **Symmetrical:** Same pattern across tyre (common)
   - **Asymmetrical:** Different inner/outer pattern (performance)
   - **Directional:** V-pattern, specific rotation direction (wet grip)
3. **Sidewall:** Connects tread to bead
   - Provides flexibility
   - Contains tyre information (more later)
   - Vulnerable to damage (cuts, bulges)
4. **Bead:** Steel wire bundles
   - Holds tyre on rim
   - Seals against wheel
   - Must fit rim exactly
5. **Carcass (body):** Internal structure
   - **Radial construction:** Cords run perpendicular to direction of travel
     - Better ride, longer life, lower rolling resistance
     - Standard on all modern cars
   - **Bias-ply construction:** Cords at angles
     - Older technology, rare now
     - Some commercial/off-road applications
6. **Belts:** Steel belts under tread (radial tyres)
   - Stabilize tread
   - Improve handling
   - Typically 2-3 layers
7. **Inner liner:** Air-tight rubber layer
   - Prevents air leakage
   - Like inner tube in tubeless design

Tyre Types:

By Use:

- **Passenger car:** Standard vehicles, balanced performance
- **SUV/Light truck:** Reinforced, higher load capacity
- **Performance:** Softer compound, better grip, faster wear
- **All-season:** Compromise for all conditions (most common)
- **Winter/Snow:** Soft compound, deep grooves for cold/snow
- **Summer:** Hard compound, shallow grooves, best dry grip
- **Run-flat:** Can drive ~80 km at 80 km/h after puncture
  - Reinforced sidewall
  - Heavier, harsher ride
  - No spare tyre needed

By Construction:

- **Tube-type:** Requires inner tube (old vehicles, some motorcycles)
- **Tubeless:** No inner tube (all modern cars)
  - Bead seals against rim
  - Safer (slow leak vs sudden deflation)

Hour 2 (10:00 AM - 11:00 AM): Tyre Information & Specifications

Reading Tyre Sidewall:

Example: 205/55R16 91H

1. Width (205):

- Section width in millimeters
- 205mm = width of tyre from sidewall to sidewall
- Wider = more grip, more fuel consumption, heavier steering

2. Aspect Ratio (55):

- Sidewall height as percentage of width
- 55 = sidewall is 55% of 205mm = 112.75mm
- Lower number = lower profile (performance)
- Higher number = taller sidewall (comfort)

3. Construction (R):

- R = Radial (standard)
- B = Bias-ply (rare)

- All modern tyres are radial

4. Rim Diameter (16):

- Wheel diameter in inches
- Must match wheel size exactly
- Common: 14", 15", 16", 17", 18", 20"

5. Load Index (91):

- Maximum load per tyre
- 91 = 615 kg per tyre
- Table converts number to kg
- Total vehicle capacity = 4 × load index
- Never use lower load index than specified

6. Speed Rating (H):

- Maximum safe sustained speed
- H = 210 km/h
- Common ratings:
  - Q = 160 km/h
  - S = 180 km/h
  - T = 190 km/h
  - H = 210 km/h
  - V = 240 km/h
  - W = 270 km/h
  - Y = 300 km/h
- Must meet or exceed vehicle's top speed

Additional Markings:

DOT Code:

- Example: DOT XXXX XXX 2319
- Last 4 digits = week and year manufactured
- 2319 = 23rd week of 2019
- Tyres have shelf life (~6 years)
- Old tyres deteriorate (rubber hardens)

M+S or M&S:

- Mud and Snow
- Basic winter capability

- Not same as true winter tyre

Snowflake Symbol (3PMSF):

- Three-Peak Mountain Snow Flake
- True winter tyre certification
- Tested winter performance

Directional Arrow:

- Shows correct rotation direction
- Critical for directional tyres
- "Rotation →"

Inside/Outside:

- Asymmetrical tyres
- "INSIDE" faces inward
- "OUTSIDE" faces outward
- Incorrect mounting = poor performance

XL or RF:

- Extra Load or Reinforced
- Higher load capacity
- Stiffer sidewall

Tread Wear Indicators:

- Small raised bars in grooves (1.6mm height)
- When tread wears to bars = replace
- Legal minimum in most countries
- Located by "TWI" or triangle marks on sidewall

Hour 3 (11:00 AM - 12:00 PM): Tyre Pressure & Balancing

Tyre Pressure:

Correct Pressure Critical For:

- Safety (braking, handling)
- Tyre life (under/over inflation causes premature wear)
- Fuel economy (under-inflation increases rolling resistance)
- Comfort

Pressure Specifications:

- Listed on door jamb sticker (driver's door)
- Also in owner's manual
- Different front and rear (sometimes)
- Different pressure for full load
- Measured in PSI (pounds per square inch) or Bar or kPa
    - 1 Bar = 14.5 PSI = 100 kPa
    - Typical: 30-35 PSI (2.0-2.4 Bar)

When to Check:

- Monthly minimum
- Before long trips
- When tyres cold (driven < 3 km)
- Hot tyres read 2-4 PSI higher (don't let air out!)

Effects of Incorrect Pressure:

Under-inflation (too low):

- Increased rolling resistance (fuel consumption)
- Edge wear on tread
- Sidewall flexing (heat buildup, possible blowout)
- Poor handling
- Faster wear

Over-inflation (too high):

- Center tread wear
- Harsh ride
- Less traction (smaller contact patch)
- More susceptible to damage (cuts, impacts)

TPMS (Tyre Pressure Monitoring System):

- Required on modern vehicles
- Sensors in each wheel
- Warns if pressure drops 25% below specification
- Dashboard warning light (!)
- Batteries in sensors eventually need replacement

Wheel Balancing:

Why Balancing Needed:

- No tyre/wheel combination perfectly balanced
- Weight distribution slightly uneven
- Heavy spots cause vibration at speed
- Vibration causes:
  - Uncomfortable ride
  - Premature tyre wear
  - Suspension component wear
  - Steering wheel shake

Static Balance:

- Weight distributed evenly around wheel
- Prevents hopping (up and down)

Dynamic Balance:

- Weight distributed evenly side-to-side
- Prevents wobbling (side to side)

Balancing Process (Awareness Level):

1. Mount tyre/wheel on balancing machine
2. Spin wheel at high speed
3. Machine detects heavy spots
4. Machine indicates where to place weights
5. Clip-on or adhesive weights added
6. Re-spin to verify
7. Repeat until balanced

When to Balance:

- With every new tyre installation
- If vibration felt at highway speeds
- After tyre rotation
- After hitting large pothole
- Annually as preventive maintenance

Wheel Weights:

- Clip-on: Traditional, on rim lip (visible)
- Adhesive: Stick to inside of wheel (hidden)
- Usually lead or zinc alloy
- Measured in grams (5g, 10g, etc.)

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Tyre Information Reading Practice

Activity 1: Sidewall Decoding (25 min)

- Students examine actual tyres
- Read and record for each tyre:
  - Width (mm)
  - Aspect ratio (%)
  - Construction (R)
  - Rim diameter (inches)
  - Load index (convert to kg using chart)
  - Speed rating (convert to km/h using chart)
  - DOT code (calculate age)
  - Any special markings (M+S, 3PMSF, XL, Directional, etc.)
- Create specification chart:

```
Tyre 1: 205/55R16 91H
- Width: 205mm
- Aspect ratio: 55%
- Construction: Radial
- Rim: 16 inches
- Load: 615 kg per tyre
- Speed: 210 km/h
- Age: 2319 = Week 23, Year 2019 (5 years old)
```

Activity 2: Tyre Age Calculation (15 min)

- Practice with multiple DOT codes
- Calculate tyre age
- Determine if replacement needed (>6 years = consider replacing)
- Understanding safety implications of old tyres

Activity 3: Matching Tyres to Vehicles (20 min)

- Given vehicle specifications
- Students find suitable replacement tyres
- Check:
  - Correct size matches
  - Load index sufficient

    - Speed rating adequate
    - Budget vs premium options
    - Tread pattern for intended use

Pass Criteria:

- ✅ Can read complete tyre specification
- ✅ Can decode DOT code for age
- ✅ Can convert load index to kg
- ✅ Can identify special markings
- ✅ Can match tyre spec to vehicle requirements

Hour 2 (2:00 PM - 3:00 PM): Tyre Inspection & Measurement

Activity 1: Tyre Pressure Check (15 min)

- Review from Day 1, with additional detail
- Students practice on multiple vehicles:
    1. Find recommended pressure on door jamb
    2. Ensure tyres cold (parked 3+ hours)
    3. Remove valve cap
    4. Press gauge firmly on valve stem
    5. Read pressure
    6. Compare to specification
    7. Adjust if needed (air compressor or release air)
    8. Replace valve cap
    9. Check all tyres including spare
- Document findings:
    - Recommended: ___ PSI
    - Actual: ___ PSI
    - Action: Added/Released ___ PSI

Activity 2: Tread Depth Measurement (25 min)

- **Why tread depth matters:**
    - Minimum legal: 1.6mm
    - Recommend replacement: 3mm
    - New tyre: 8-10mm typically
    - Below minimum = illegal, dangerous (especially wet)

Measurement methods:

1. Tread depth gauge:

- Most accurate method
- Probe extends into groove
- Read measurement on scale
- Check multiple points across tread
- Check all four tyres

2. Tread wear indicators:

- Built-in bars at 1.6mm
- When tread flush with bars = replace
- Located by TWI marking on sidewall

3. 1-Rupee coin test:

- Insert coin into groove
- If can see above marking = tread too worn
- Quick check method

Measurement procedure:

- Measure at 3 points across tread width:
  - Inside edge
  - Center
  - Outside edge
- Measure at multiple locations around tyre
- Record minimum depth found
- Uneven depth = alignment or pressure issue

Activity 3: Visual Tyre Inspection (20 min)

- **Systematic inspection:**

Tread surface:

- Wear pattern (even, center, edges, feathering)
- Cuts, punctures, embedded objects
- Bulges or blisters (internal damage)
- Exposed cords/belts (immediate replacement!)

Sidewalls:

- Cuts, cracks, gouges
- Bulges (impact damage, weak spot)

- Weather checking (age-related cracking)
- Scuffing (rubbing on curbs)

Bead area:

- Damage to bead
- Corrosion on wheel rim
- Proper seating on rim

Valve stem:

- Cracks in rubber
- Bent or damaged
- Cap present (keeps dirt out)
- Leaking (soapy water test)

Foreign objects:

- Nails, screws, glass
- Determine if puncture repairable:
  - Tread area only (not sidewall)
  - < 6mm diameter
  - Not too close to previous repair
- Create inspection report for each tyre
- Recommend: OK, Monitor, Replace

Pass Criteria:

- ✅ Can check and adjust tyre pressure correctly
- ✅ Can measure tread depth with gauge
- ✅ Can identify tread wear indicators
- ✅ Performs thorough visual inspection
- ✅ Recognizes tyres needing replacement
- ✅ Documents findings professionally

Hour 3 (3:00 PM - 4:00 PM): Wheel Inspection & Balancing Awareness

Activity 1: Wheel Inspection (20 min)

- **Steel wheels:**
  - Check for bends, dents
  - Check for rust perforation

    - Check for damage around lug nut holes
    - Relatively durable, ugly when damaged
- **Alloy wheels:**
    - Check for cracks (safety critical!)
    - Check for curb rash (cosmetic but can progress)
    - Check for bends
    - Check for corrosion (white powder)
    - More attractive, lighter, but more fragile
- **General wheel checks:**
    - Run hand around inner rim (sharp edges = damage)
    - Check bead seating surface (smooth, no corrosion)
    - Check center bore (hub-centric fitment)
    - Check lug nut seats (conical vs flat)
    - Look for cracks starting from lug holes

Activity 2: Wheel Balance Demonstration (20 min)

- Instructor demonstrates balancing machine
- Students observe process:

Balancing procedure:

1. Clean wheel thoroughly (remove old weights, dirt)
2. Mount wheel on balancing machine cone
3. Secure with nut
4. Lower hood (safety)
5. Spin wheel at speed
6. Machine measures imbalance
7. Machine stops, indicates weight locations:
    - Inside and outside positions
    - Weight amount needed (grams)
8. Install weights at indicated positions
9. Re-spin to verify balance
10. Adjust if needed

Understanding balance readings:

- Machine shows grams needed at clock positions
- Example: "40g at 3 o'clock, 25g at 10 o'clock"
- When reading 0g = balanced
- Tolerance: Typically ±5g acceptable

Types of imbalance:

- **Static:** Heavy spot on one side
- **Dynamic:** Weight distributed unevenly across width
- Modern machines correct both simultaneously

Activity 3: Vibration Diagnosis (20 min)

- **Understanding vibration sources:**

Speed-dependent vibration:

- **40-60 km/h:** Usually wheel balance
- **80-120 km/h:** Could be balance or alignment
- **All speeds:** Likely suspension or drivetrain

Wheel-related vibration symptoms:

- **Steering wheel shake:** Front wheel balance issue
- **Seat/floor shake:** Rear wheel balance issue
- **Constant frequency:** Wheel balance
- **Varies with braking:** Warped rotor
- **Varies with turning:** Wheel bearing or CV joint

Practice scenarios:

- Instructor describes vibration symptoms
- Students diagnose likely cause:
  - "Steering wheel shakes at 80 km/h"
  - "Vibration when braking"
  - "Constant vibration regardless of speed"
  - "Clicking noise when turning"

Pass Criteria:

- ✅ Can inspect wheels for damage
- ✅ Understands wheel balancing process
- ✅ Can interpret balance machine readings
- ✅ Can diagnose vibration sources
- ✅ Knows when balancing is needed

DAY 16 ASSESSMENT:

Practical Skills Checklist:

- [ ] Reads tyre sidewall specifications completely
- [ ] Decodes DOT code for age
- [ ] Checks tyre pressure correctly
- [ ] Measures tread depth accurately
- [ ] Performs visual tyre inspection
- [ ] Inspects wheels for damage
- [ ] Understands balancing process
- [ ] Recognizes tyres needing replacement

Knowledge Check (Verbal):

- What does 205/55R16 mean?
- What is legal minimum tread depth?
- What causes center tread wear?
- When should tyres be balanced?
- What does the snowflake symbol mean?

DAY 17: COMFORT & SAFETY SYSTEMS + INSPECTIONS

Learning Objectives:

- Understand HVAC system basics
- Identify safety system components
- Understand airbag precautions
- Perform safety system visual inspection

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): HVAC System Fundamentals

Purpose of HVAC:

- **Heating:** Warm cabin in cold weather
- **Ventilation:** Fresh air circulation
- **Air Conditioning:** Cool cabin in hot weather
- **Defogging/Defrosting:** Clear windows for visibility

Heating System:

- Uses engine heat (waste heat)
- **Components:**
  - **Heater core:** Small radiator inside dashboard
  - **Heater hoses:** Hot coolant from engine
  - **Blend door:** Controls air flow through heater core
  - **Blower motor:** Fan that moves air
  - **Temperature control:** Driver adjusts blend door

Operation:

1. Hot coolant from engine flows to heater core
2. Blower motor blows air through heater core
3. Hot air enters cabin through vents
4. Temperature control mixes hot and cold air

No heat problems:

- Low coolant level
- Stuck thermostat (open)
- Clogged heater core
- Blend door stuck
- Air in system

Air Conditioning System:

- Uses refrigeration cycle
- **Components:**
  - **Compressor:** Pumps refrigerant, driven by engine belt
  - **Condenser:** Radiator in front, releases heat
  - **Receiver/drier:** Removes moisture
  - **Expansion valve/orifice tube:** Restricts flow
  - **Evaporator:** Radiator inside dashboard, absorbs heat
  - **Refrigerant:** R134a (older) or R1234yf (new, expensive)
  - **Blower motor:** Same as heating

Refrigeration cycle:

1. Compressor compresses refrigerant (hot, high pressure gas)
2. Condenser cools refrigerant (becomes liquid)
3. Expansion valve restricts flow (pressure/temp drop)
4. Evaporator absorbs cabin heat (refrigerant becomes cold gas)
5. Back to compressor (cycle repeats)

A/C operation:

- Press A/C button
- Compressor clutch engages
- Refrigerant circulates
- Cold air from evaporator
- Blower motor delivers to cabin
- Drain tube removes condensation water

No cooling problems:

- Low refrigerant (leak)
- Compressor failure
- Clogged condenser
- Electrical issues
- Blower motor failure

Hour 2 (10:00 AM - 11:00 AM): Cabin Air & Ventilation

Cabin Air Filter:

- Filters air entering cabin
- **Located:** Behind glove box or under hood
- **Function:**
  - Removes dust, pollen, pollution
  - Protects occupants
  - Keeps evaporator clean
- **Types:**
  - Paper (standard)
  - Activated carbon (removes odors)
  - HEPA (hospital-grade filtration)
- **Replacement:** Every 10,000-20,000 km or annually
- **Symptoms of clogged filter:**
  - Reduced airflow
  - Musty smell
  - Fogging windows
  - Poor A/C cooling

Ventilation System:

- Brings fresh outside air

- **Fresh air mode:** Outside air enters
- **Recirculation mode:** Cabin air circulates
- **When to recirculate:**
  - Maximum A/C cooling (short term)
  - Dusty environment outside
  - Tunnel driving
- **When to use fresh air:**
  - Normal driving (prevents CO_2 buildup)
  - Window defogging
  - Odor elimination

Blower Motor & Fan Speeds:

- Electric motor drives fan
- Multiple speeds (typically 4-5)
- **Auto mode (climate control):**
  - ECU controls fan speed automatically
  - Maintains set temperature
  - Adjusts based on conditions

Windshield Defog/Defrost:

- Uses A/C + heater combination
- **Defog (front):**
  - A/C removes moisture
  - Warm air prevents condensation
  - Maximum airflow to windshield
- **Defrost (rear):**
  - Electric heating elements in glass
  - Draws significant current
  - Turns off automatically (timer)

Hour 3 (11:00 AM - 12:00 PM): Safety Systems

Seat Belts:

- **Most important safety device**
- Reduces injury 50-70% in crash
- **Components:**
  - Webbing (strong fabric)
  - Buckle mechanism

    - Retractor (spring-loaded reel)
    - Pre-tensioner (pulls belt tight in crash)
    - Load limiter (releases slightly to prevent injury)

Seat Belt Operation:

- Pulls out smoothly for use
- Locks when:
    - Pulled quickly
    - Vehicle decelerates suddenly
    - Inertial reel mechanism
- Pre-tensioner fires once (crash only)
- Must be replaced after accident

Seat Belt Inspection:

- Check webbing for fraying, cuts
- Check smooth operation (locks when jerked)
- Check buckle latches securely
- Never modify or add slack

Airbag System (SRS - Supplemental Restraint System):

- **"Supplemental"** = works with seat belts, not replacement
- Deploys in moderate to severe frontal crash
- Inflates in 20-30 milliseconds (faster than blink)
- Nitrogen gas inflates bag
- Deflates immediately after impact

Airbag Components:

- **Airbag modules:**
    - Driver airbag (steering wheel)
    - Passenger airbag (dashboard)
    - Side airbags (seat/door)
    - Curtain airbags (roof rail)
    - Knee airbag (some vehicles)
- **Crash sensors:**
    - Detect impact severity
    - Multiple sensors (front, sides)
- **SRS ECU:**
    - Evaluates sensor data
    - Decides whether to deploy

    - Which bags to deploy
- **Clock spring:**
    - Maintains electrical connection to steering wheel airbag
    - Allows steering wheel rotation

SRS Warning Light:

- Illuminates briefly at startup (system check)
- Should turn off after few seconds
- **Stays on = system fault:**
    - Crash sensor issue
    - Wiring problem
    - Airbag module fault
    - Clock spring failure
- Must be diagnosed immediately (airbags may not work)

CRITICAL SAFETY RULES WHEN WORKING ON VEHICLES:

1. Disconnect Battery FIRST:

- Wait 10 minutes after disconnecting (capacitors discharge)
- Prevents accidental deployment
- Deployment can cause serious injury

2. Never Probe Airbag Connectors:

- Can cause deployment
- Use manufacturer procedures only

3. Yellow Connectors = Airbag:

- Do not disconnect without proper procedure
- Often have safety clips

4. Store Airbags Safely:

- Face up (deployment side up)
- Never place objects on top
- Away from heat sources
- In original packaging if possible

5. Never Modify Airbag System:

- No aftermarket steering wheels without airbag
- No seat covers that block side airbags

- No stickers on airbag covers

6. Scan for Codes After Service:

- Any work near airbags
- Battery disconnection
- Must verify system operational

Occupant Safety Features:

Crumple Zones:

- Front and rear structures designed to deform
- Absorb impact energy
- Protect passenger compartment (safety cell)

Safety Cell:

- Rigid passenger compartment
- Reinforced pillars (A, B, C)
- Strong roof structure

Side Impact Beams:

- Steel bars in doors
- Resist side impacts
- Protect occupants

Child Safety:

- ISOFIX/LATCH anchor points
- Top tether anchors
- Child seat installation points
- Rear door child locks

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): HVAC System Identification & Inspection

Activity 1: HVAC Component Location (30 min)

- Identify on actual vehicle:

Under hood:

- **A/C compressor:** Belt-driven, usually on side of engine
  - Magnetic clutch visible
  - Electrical connector
  - High/low pressure lines
- **Condenser:** Radiator in front of engine radiator
  - Aluminum, thin fins
  - Lines from compressor
- **Receiver/drier:** Canister on condenser or line
  - Contains desiccant (moisture absorber)
- **Heater hoses:** Two hoses to firewall
  - Hot coolant in/out
  - Go to heater core inside

Inside cabin:

- **Blower motor:** Under dashboard (passenger side usually)
- **Cabin air filter access:** Behind glove box or under hood
- **Evaporator:** Behind dashboard (not visible)
- **Temperature controls:** Dashboard
- **Vents:** Dashboard, floor, defrost
- Draw HVAC system diagram
- Trace airflow path
- Note component locations

Activity 2: Cabin Air Filter Inspection/Replacement (30 min)

- **Access cabin filter:**
  - Most common: Behind glove box
    1. Remove glove box screws/clips
    2. Lower glove box
    3. Remove filter cover
    4. Slide out old filter
  - Some: Under hood near windshield
    1. Remove cowl panel
    2. Access filter housing
    3. Remove old filter
- **Inspect old filter:**
  - Dirt/debris accumulation?
  - Blocked airflow?
  - Odor?
  - Leaves, insects?

- **Install new filter:**
  - Note arrow showing airflow direction
  - Slide into housing correctly
  - Secure cover
  - Reinstall glove box/panel
- **Test airflow:**
  - Turn on blower (all speeds)
  - Should feel improved airflow with new filter

Pass Criteria:

- ✅ Can locate A/C compressor and condenser
- ✅ Can identify heater hoses
- ✅ Can access and replace cabin air filter
- ✅ Understands proper filter orientation
- ✅ Creates HVAC system diagram

Hour 2 (2:00 PM - 3:00 PM): HVAC Functional Checks

Activity 1: A/C System Operation Check (20 min)

- **Basic functionality test:**
  1. Start engine
  2. Turn on A/C (max cold, max fan)
  3. **Observe compressor clutch:**
     - Should engage (click sound)
     - Center of compressor pulley rotates
     - If doesn't engage = electrical issue or low refrigerant
  4. **Check airflow:**
     - Strong from vents
     - Cold air (should be <10°C at vent)
  5. **Check condenser fan:**
     - Should run when A/C on
     - Electric fan cycles on/off
  6. **Check drain tube:**
     - Under vehicle, passenger side
     - Water should drip (condensation = system working)
     - No water = possible evaporator issue
- **Common issues observed:**
  - Compressor doesn't engage

    - Air not cold
    - Weak airflow
    - Unusual noises
    - Bad smell (musty = dirty evaporator)

Activity 2: Heating System Check (15 min)

- **Heat functionality test:**
  1. Engine warm (operating temperature)
  2. Turn heat to maximum hot
  3. Turn fan to medium speed
  4. **Check output:**
     - Should feel hot air from vents
     - Temperature rises within 1-2 minutes
  5. **Check all vent positions:**
     - Face vents
     - Floor vents
     - Defrost vents
     - Mixed mode
  6. **Check temperature control:**
     - Vary from cold to hot
     - Should feel temperature change
- **No heat problems:**
  - Check coolant level first
  - Check heater hoses hot (both should be)
  - Check blend door operation

Activity 3: Blower Motor & Controls Check (25 min)

- **Test all fan speeds:**
  - Speed 1 (low)
  - Speed 2
  - Speed 3
  - Speed 4 (high)
  - Each should be noticeably different
  - All speeds should work
- **Abnormal conditions:**
  - Only high speed works = resistor failure
  - No speeds work = motor or fuse failure
  - Noise at certain speed = motor bearing
- **Recirculation vs fresh air:**

    - Test both modes
    - Recirculation: Door closes outside air
    - Fresh air: Outside air enters
    - Should hear door operation
- **Defrost/defog test:**
    - Turn on defrost mode
    - Air should direct to windshield
    - A/C compressor usually activates automatically
    - Rear defrost button (if equipped) - check rear window clears

Pass Criteria:

- ✅ Can verify A/C compressor operation
- ✅ Can check cold air output
- ✅ Can test heating system
- ✅ Can verify all blower speeds
- ✅ Can test all vent positions
- ✅ Documents findings

Hour 3 (3:00 PM - 4:00 PM): Safety System Inspection

Activity 1: Seat Belt Inspection (20 min)

- **Visual inspection all seat belts:**
    - Pull out fully, inspect entire webbing
    - Look for:
        - Fraying, cuts, burns
        - Twisted webbing
        - Fading (UV damage weakens)
        - Stains (may indicate contamination)
- **Functional test:**
    - Pull slowly - should extend smoothly
    - Pull quickly - should lock (inertial lock)
    - Tilt vehicle - should lock (pendulum lock)
    - Release - should retract fully and smoothly
    - Buckle should latch securely
    - Buckle release button should work
- **Pre-tensioner check:**
    - **DO NOT TEST (deploys once)**
    - Visual check only:

    - Look for damage to mechanism
    - Check electrical connector secure
    - Check SRS light not on
- Document condition of all belts
- Any belt with damage = replace (safety critical!)

Activity 2: Airbag System Visual Inspection (20 min)

CRITICAL SAFETY REMINDER:

- DO NOT disconnect any yellow connectors
- DO NOT probe airbag components
- Visual inspection only
- Battery remains connected for this inspection

Inspection points:

- **SRS warning light test:**
    - Turn key to ON (engine off)
    - Light should illuminate briefly
    - Should turn off after 5-10 seconds
    - Stays on = fault (requires scan tool diagnosis)
- **Steering wheel airbag:**
    - Center cover should be intact
    - No cracks in cover
    - Horn works (shares electrical path)
    - No aftermarket steering wheel
- **Passenger airbag:**
    - Dashboard cover intact
    - No cracks or damage
    - Passenger airbag ON/OFF switch (if equipped)
- **Side airbag indicators (if equipped):**
    - Look for "AIRBAG" or "SRS" tags on seats
    - Check seat covers don't block deployment
- **B-pillar tags:**
    - Yellow warning stickers
    - Indicate side airbags present
- **Crash sensors:**
    - Usually not visible
    - If visible (some in bumper), check not damaged

Activity 3: Structural Safety Inspection (20 min)

- **Occupant safety features:**

Body structure:

- Look for previous accident repair:
  - Paint overspray
  - Misaligned panels
  - Replaced parts
  - Welding evidence
- Check A, B, C pillar condition
  - No bends or damage
  - Important for rollover protection

Doors:

- Check side impact beams (not visible, but note presence)
- Door closes properly (alignment)
- Latches secure
- Child safety locks work (rear doors)

ISOFIX/LATCH points:

- Located between seat cushions (rear)
- Check not damaged or obstructed
- Top tether anchor (behind seat)

Glass:

- Windshield chips/cracks (vision obstruction)
- Delamination (bubbling)
- Damage near airbag deployment zone
- Create safety system inspection report
- Note any concerns
- Understand these are visual checks only
- Advanced diagnostics require scan tools

Pass Criteria:

- ✅ Can inspect seat belts thoroughly
- ✅ Can verify seat belt operation
- ✅ Can check SRS warning light
- ✅ Can visually inspect airbag covers
- ✅ Identifies safety concerns
- ✅ Follows safety precautions with airbags

- ✅ Documents findings properly

DAY 17 ASSESSMENT:

Practical Skills Checklist:

- [ ] Locates HVAC components
- [ ] Replaces cabin air filter
- [ ] Tests A/C and heating operation
- [ ] Checks all blower speeds
- [ ] Inspects seat belts comprehensively
- [ ] Checks SRS warning light
- [ ] Performs safety system visual inspection
- [ ] Follows airbag safety precautions

Knowledge Check (Verbal):

- How does the heater system work?
- Name three main A/C system components
- What does the cabin air filter do?
- Why must we wait 10 minutes after disconnecting battery near airbags?
- What does SRS warning light staying on indicate?

DAY 18: CHASSIS INTEGRATION + COMPREHENSIVE INSPECTION

Learning Objectives:

- Integrate all chassis system knowledge
- Understand system interactions
- Perform comprehensive pre-purchase inspection
- Create professional vehicle assessment report

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Chassis Systems Integration

How Chassis Systems Work Together:

Complete Vehicle Control Flow:

1. **Driver inputs steering** → Steering system
2. **Suspension maintains tire contact** → Proper steering response
3. **Tyres provide grip** → Steering and braking possible
4. **Brakes provide stopping power** → Weight transfers forward (suspension compresses)
5. **All systems must coordinate** for safe, controlled driving

System Interdependencies:

Suspension ↔ Steering:

- Suspension geometry affects steering response
- Worn suspension components cause:
  - Poor steering return
  - Wandering
  - Uneven tire wear (then affects grip)
- Steering loads transmitted through suspension
- Ball joints serve both systems

Suspension ↔ Tyres:

- Suspension maintains tire contact patch
- Worn shocks = inconsistent contact = uneven tire wear
- Improper alignment (suspension geometry) = rapid tire wear
- Tire pressure affects suspension behavior:
  - Low pressure = softer ride, but poor handling
  - High pressure = harsh ride, less grip

Tyres ↔ Brakes:

- Brakes can only work as well as tire grip allows
- Worn tires = longer stopping distance
- Proper tire pressure critical for braking
- ABS relies on tire grip to function
- Uneven tire wear causes:
  - Pulsating braking
  - Pull to one side

Steering ↔ Brakes:

- Braking causes weight transfer (dive)
- Affects steering response during braking
- Brake pull affects steering (vehicle pulls)

- Must be able to steer while braking (ABS maintains this)

All Systems → Safety:

- Any single system failure reduces safety
- Multiple system problems = dangerous
- Regular inspection/maintenance essential
- Preventive maintenance prevents cascading failures

Example Scenario: Emergency Maneuver

1. Driver sees obstacle, brakes hard and swerves
2. **Tyres** must provide grip for both braking and cornering
3. **Brakes** must stop vehicle without locking (ABS)
4. **Suspension** must keep tires planted through maneuver
5. **Steering** must respond precisely and predictably
6. **All systems working together** = safe outcome
7. **One system failing** (bald tires, worn shocks, etc.) = crash

---

Hour 2 (10:00 AM - 11:00 AM): Vehicle Inspection Standards

Why Comprehensive Inspections Matter:

- **Pre-purchase:** Avoid buying problem vehicle
- **Periodic:** Catch problems early
- **Pre-trip:** Safety for long journeys
- **Post-accident:** Identify hidden damage
- **Maintenance:** Professional service

Types of Inspections:

1. Safety Inspection (Quick Check):

- Focus on safety-critical items
- 15-20 minutes
- Before every drive ideally:
  - Lights
  - Tires (visual, pressure)
  - Brakes (basic test)
  - Fluids (oil, coolant)
  - Warning lights

2. Routine Service Inspection:

- Part of scheduled maintenance
- 30-45 minutes
- Comprehensive checks:
  - All fluids
  - All filters
  - Belts and hoses
  - Brake system
  - Suspension components
  - Tire condition and wear
  - Battery
  - Exhaust system

3. Pre-Purchase Inspection:

- Before buying used vehicle
- 60-90 minutes
- Most thorough:
  - All routine items plus:
  - Test drive evaluation
  - Paint thickness (accident history)
  - Chassis rust and damage
  - Evidence of poor repairs
  - Functionality of all systems
  - Diagnostic scan (fault codes)
  - Compression test (if concerned)

4. Annual Safety Inspection (PUC + Fitness):

- Legal requirement in many states
- Emissions testing
- Safety component verification
- Lighting and visibility
- Brake effectiveness testing
- Certificate issued if passed

Inspection Documentation:

- **Critical for professional work:**
  - Protects shop legally
  - Shows customer what's needed

    - Tracks vehicle history
    - Justifies recommendations

Inspection Report Format:

- Vehicle information (make, model, VIN, mileage)
- Date and inspector name
- Check items with status:
    - ✅ Good/Pass
    - ⚠️ Monitor/Advisory
    - ❌ Fail/Immediate attention
- Photos of problems
- Recommendations prioritized:
    - **Red (Urgent):** Safety critical, illegal, breakdown imminent
    - **Amber (Soon):** Will worsen, do within 1 month
    - **Green (Monitor):** Watch, check next service
- Cost estimates
- Customer signature

Hour 3 (11:00 AM - 12:00 PM): Professional Standards & Communication

Professional Inspector Qualities:

- **Thorough:** Don't miss anything
- **Systematic:** Use checklist, same order every time
- **Honest:** Report what you find, not what customer wants to hear
- **Knowledgeable:** Understand what you're looking at
- **Communicative:** Explain findings clearly

Inspection Ethics:

- Never recommend unnecessary work
- Never overlook problems (liability!)
- Be honest about severity
- Provide options when appropriate
- Disclose conflicts of interest
- Maintain customer privacy

Communicating Findings:

To Customers (Non-Technical):

- Use simple language
- Show the problem (photos, physical inspection)
- Explain consequences of not fixing
- Provide cost estimates
- Answer questions patiently
- Never use scare tactics
- Build trust through honesty

Example - Good Communication: "Your front brake pads are down to 2mm. The minimum safe thickness is 3mm, and the legal minimum is 1.6mm. If you keep driving, you'll hear metal-on-metal grinding within a few hundred kilometers, and that will damage your rotors, turning a ₹2,000 repair into a ₹5,000 repair. I recommend replacing them this week."

Example - Poor Communication: "Your pads are shot. Rotors are gonna warp if you don't fix it now. Probably need calipers too. This is dangerous."

To Other Technicians (Technical):

- Use proper terminology
- Be specific with measurements
- Reference locations precisely
- Include relevant specs
- Note any unusual conditions

Professional Documentation:

- Clear, legible writing or typing
- Accurate measurements
- Good quality photos
- Organized format
- Signed and dated
- Customer copy provided

Building Customer Trust:

- Show them what you found
- Explain why it matters
- Provide evidence (measurements, photos)
- Give them options
- Respect their budget
- Follow up on recommendations
- Warranty your work

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Comprehensive Vehicle Walk-Around

Activity 1: Systematic Exterior Inspection (30 min)

- **Inspection sequence (always same order):**

1. Front:

- Overall condition, accident damage?
- Headlights (lenses clear, bulbs work)
- Grille (intact, no missing pieces)
- Bumper (cracks, mounting secure)
- Windshield (chips, cracks)
- Hood (opens/closes properly, aligned)
- Hood release (works smoothly)

2. Front Left:

- Fender (dents, rust, alignment)
- Door (opens/closes, aligned, rust)
- Window (operates smoothly)
- Mirror (intact, adjusts)
- Side marker light

3. Rear Left:

- Same door/mirror checks
- Rear fender/quarter panel
- Fuel filler door (opens, cap present)
- Taillights (lenses, bulbs)

4. Rear:

- Bumper condition
- Trunk/hatch (opens, stays open, aligned)
- Reverse lights, license plate light
- Exhaust pipe (rust, hanging properly)

5. Rear Right & Front Right:

- Mirror checks of left side

6. Overall:

- Paint condition (faded, peeling)
- Panel gaps (even = good, uneven = poor repair)
- Body damage evidence
- Rust (especially wheel arches, rocker panels)
- Underbody (if visible, lift not available yet)
- Create exterior condition report
- Note any damage with photos
- Mark on vehicle diagram

Activity 2: Under-Hood Inspection (30 min)

- **Systematic under-hood check:**

Fluid Levels (review from Week 2):

- Engine oil
- Coolant
- Brake fluid
- Power steering fluid
- Windshield washer

Visual Component Check:

- Battery (condition, terminals)
- Belts (cracks, tension)
- Hoses (cracks, bulges, clamps)
- Air filter (condition)
- Fluid leaks (colors indicate source)
- Wiring (damaged insulation)
- Emission components (connected)
- Engine mounts (broken, leaking)
- Exhaust manifold (cracks)

Listen for issues:

- Squealing (belt)
- Clicking (valve train)
- Knocking (bearings)
- Hissing (leaks)
- Document all findings
- Rate each item (Good/Monitor/Fail)

Pass Criteria:

- ✅ Conducts systematic exterior inspection
- ✅ Identifies panel damage and misalignment
- ✅ Checks all lights and glass
- ✅ Performs comprehensive under-hood inspection
- ✅ Documents findings with photos

Hour 2 (2:00 PM - 3:00 PM): Under-Vehicle & Chassis Inspection

Activity 1: Lifting and Safety Setup (10 min)

- Lift vehicle safely (practiced extensively)
- Position lighting
- Prepare inspection tools
- Safety glasses on (dirt, rust falls)

Activity 2: Complete Underside Inspection (50 min)

- **Inspection sequence (front to rear):**

Suspension (Front and Rear):

- Shocks/struts (leaks)
- Springs (broken coils)
- Control arms (bent, bushings cracked)
- Ball joints (boots torn, play)
- Sway bar links (play, noise)
- Tie rods (boots, play)
- CV boots (torn, leaking)

Steering:

- Steering rack (leaks)
- Rack boots (torn)
- Power steering hoses (leaks)
- Steering linkage (play)

Brakes:

- Brake lines (rust, leaks)
- Brake hoses (cracks, bulges)
- Calipers (leaks, seized pins)
- Rotors (grooves, cracks)

- Pads (thickness)
- Drums (if present, condition)

Drivetrain:

- Engine oil pan (leaks, damage)
- Transmission (leaks)
- Differential (leaks if RWD/AWD)
- Driveshaft (if present, U-joint wear)
- Axle seals (leaks)

Exhaust:

- Entire exhaust system (rust, holes)
- Hangers (broken)
- Heat shields (loose, missing)
- Catalytic converter (physical damage)

Structure:

- Frame/unibody (rust, damage)
- Floor pans (rust perforation)
- Subframes (rust, bushings)
- Mounting points (cracks)

Fluid Drips:

- Look for active leaks
- Check pavement under vehicle
- Identify fluid by color/location
- Create comprehensive underside report
- Use flashlight to see into crevices
- Take photos of any concerns
- Measure brake pad thickness
- Check for play in suspension/steering

Pass Criteria:

- ✅ Lifts vehicle safely
- ✅ Conducts systematic underside inspection
- ✅ Identifies leaks and damage
- ✅ Checks all chassis components
- ✅ Measures brake pad thickness
- ✅ Documents findings thoroughly

Hour 3 (3:00 PM - 4:00 PM): Final Assessment & Report Creation

Activity 1: Interior & Electrical Check (20 min)

- **Interior condition:**
  - Seat belts (all positions)
  - SRS warning light
  - Instrument cluster (all gauges work)
  - All switches and controls (windows, locks, mirrors)
  - HVAC (all functions)
  - Infotainment (if present)
  - Odometer reading (verify with service records)
- **Electrical:**
  - Battery voltage (12.4-12.8V off, 13.5-14.5V running)
  - All lights function
  - Horn works
  - Wipers (all speeds)
  - Washer spray

Activity 2: Test Drive Evaluation (If Permitted) (20 min)

- **NOTE:** Fundamentals students observe only
- **Test drive checklist:**

Starting:

- Engine starts smoothly
- No unusual noises
- Warning lights go out

Acceleration:

- Smooth power delivery
- No hesitation
- Transmission shifts properly
- No unusual vibrations

Cruising:

- Straight-line tracking
- No pulling
- No vibrations at speed

- Engine runs smoothly

Braking:

- Straight stops
- No pulling
- No vibrations
- ABS test (safe location only)

Steering:

- Responsive
- Returns to center
- No play
- No noises

Suspension:

- Comfortable ride
- No excessive bouncing
- No clunks or rattles

Activity 3: Comprehensive Report Creation (20 min)

- **Create professional inspection report:**

Header:

- Date, inspector name
- Vehicle details (make, model, year, VIN, mileage)

Executive Summary:

- Overall condition rating
- Major findings summary
- Recommendation (buy/don't buy if pre-purchase)

Detailed Findings by System:

- **Exterior:** Condition, damage, rust
- **Engine:** Leaks, condition, performance
- **Transmission:** Operation, leaks
- **Suspension:** Component condition, wear
- **Steering:** Play, leaks, alignment needed?
- **Brakes:** Pad life remaining, rotor condition

- **Tires:** Tread depth, wear patterns, age
- **Electrical:** Battery, lights, functions
- **HVAC:** Cooling, heating, blower
- **Safety:** Seat belts, airbags, structural
- **Exhaust/Emissions:** Leaks, condition

Recommendations (Prioritized):

- 🔴 **Urgent (Red):**
  - Safety critical issues
  - Illegal to drive
  - Will cause breakdown soon
  - Example: "Brake pads 1mm, metal-to-metal imminent"
- 🟡 **Soon (Amber):**
  - Will worsen if ignored
  - Do within 1 month
  - Example: "Rear shocks leaking, ride quality deteriorating"
- 🟢 **Monitor (Green):**
  - Watch at next service
  - Not critical yet
  - Example: "Tires 4mm tread, good for 10,000 km more"

Cost Estimates:

- Parts and labor for each item
- Total for urgent repairs
- Total for all recommended work

Photos:

- Attach photos of any issues
- Label clearly

Sign-Off:

- Inspector signature
- Date
- Customer signature (acknowledges receipt)

Pass Criteria:

- ✅ Completes comprehensive interior check
- ✅ Understands test drive evaluation criteria
- ✅ Creates professional inspection report

- ✅ Prioritizes recommendations correctly
- ✅ Provides clear cost estimates
- ✅ Attaches supporting photos
- ✅ Communicates findings clearly

DAY 18 ASSESSMENT:

Comprehensive Week 3 Evaluation:

Practical Skills (All Week 3 Content):

- [ ] Identifies all suspension components
- [ ] Identifies all steering components
- [ ] Identifies all brake components
- [ ] Reads tire specifications completely
- [ ] Performs suspension inspection
- [ ] Checks steering play and operation
- [ ] Measures brake pad thickness
- [ ] Measures tire tread depth
- [ ] Checks tire pressure
- [ ] Inspects seat belts and SRS
- [ ] Tests HVAC functions
- [ ] Conducts comprehensive vehicle inspection
- [ ] Creates professional inspection report
- [ ] Communicates findings to customers

Knowledge Check (Written - 30 questions):

Suspension (5 questions):

1. Name three suspension types
2. What does a shock absorber do?
3. What causes body roll?
4. Name three suspension components
5. What is MacPherson strut?

Steering (5 questions): 6. Two types of power steering? 7. What connects rack to wheel? 8. Typical steering free play? 9. What does EPS stand for? 10. Name three causes of steering pull

Brakes (5 questions): 11. What is hydraulic multiplication? 12. Disc vs drum brake difference? 13. Minimum brake pad thickness? 14. How does ABS work? 15. What causes

brake pulsation?

Tires (5 questions): 16. What does 205/55R16 mean? 17. Legal minimum tread depth? 18. What causes center tread wear? 19. How often balance tires? 20. What is TWI marking?

Safety (5 questions): 21. What does SRS stand for? 22. Why wait after disconnecting battery? 23. What are yellow connectors? 24. How do seat belt pre-tensioners work? 25. What is ISOFIX?

Integration (5 questions): 26. How do suspension and tires interact? 27. Why inspect all systems together? 28. Name three levels of inspection 29. What is pre-purchase inspection? 30. How to prioritize repair recommendations?

Pass Mark: 70% (21/30)

WEEK 3 SUMMARY & OUTCOMES

Week 3 Learning Journey: "Mastering Vehicle Control and Safety Systems"

Day 13: Suspension systems, wheel handling, torque procedures **Day 14:** Steering systems, alignment awareness, wear patterns **Day 15:** Brake systems, ABS, brake inspection and measurement **Day 16:** Tire construction, specifications, inspection, balancing **Day 17:** HVAC systems, safety features, airbag precautions **Day 18:** System integration, comprehensive inspection, professional reporting

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Week 3 Skills Achieved:

✅ **Suspension Skills:**

- Identifies suspension types and components
- Performs bounce test
- Inspects shocks, springs, bushings
- Recognizes worn suspension components

✅ **Steering Skills:**

- Identifies steering system types
- Locates steering components
- Measures steering play
- Inspects tie rods and boots
- Checks PS fluid

✅ Brake Skills:

- Identifies disc and drum brakes
- Measures brake pad thickness
- Inspects rotors
- Checks brake hoses and lines
- Tests parking brake
- Recognizes brake problems

✅ Tire Skills:

- Reads tire specifications
- Decodes DOT codes
- Checks tire pressure correctly
- Measures tread depth
- Performs visual tire inspection
- Identifies wear patterns
- Understands balancing

✅ Comfort/Safety Skills:

- Locates HVAC components
- Replaces cabin air filter
- Tests A/C and heating
- Inspects seat belts
- Checks SRS warning light
- Follows airbag safety procedures

✅ Integration Skills:

- Performs comprehensive inspections
- Creates professional reports
- Prioritizes recommendations
- Communicates findings clearly
- Understands system interactions